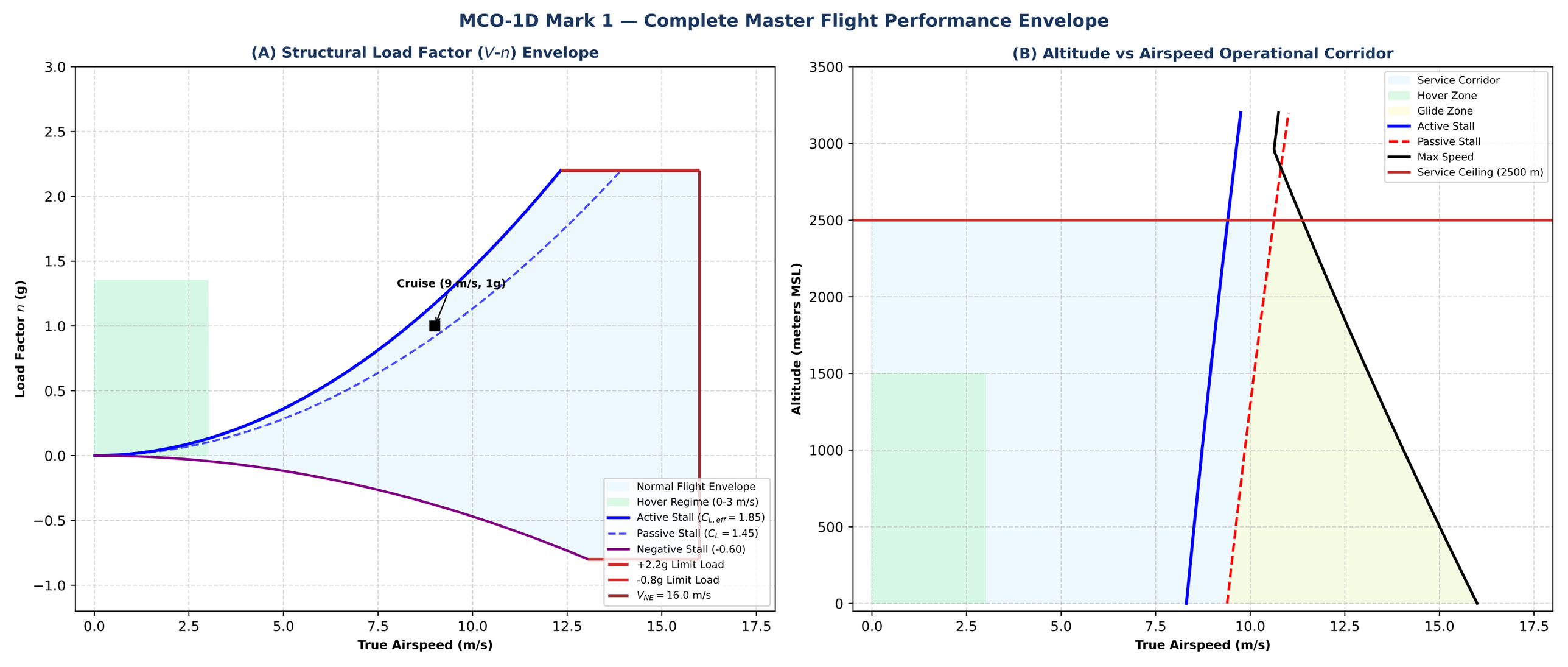
MCO-1D Mark 1 — Complete Master Flight Performance Envelope
(A) Structural Load Factor (V-n) Envelope
Load Factor n (g)
True Airspeed (m/s)
Cruise (9 m/s, 1g)
Normal Flight Envelope
Hover Regime (0-3 m/s)
Active Stall (C_L,eff = 1.85)
Passive Stall (C_L = 1.45)
Negative Stall (-0.60)
+2.2g Limit Load
-0.8g Limit Load
V_NE = 16.0 m/s
(B) Altitude vs Airspeed Operational Corridor
Altitude (meters MSL)
True Airspeed (m/s)
Service Corridor
Hover Zone
Glide Zone
Active Stall
Passive Stall
Max Speed
Service Ceiling (2500 m)